PART II: PATTERNS

Topic 10: Noise and Irregular Fluctuations

The Core Concept

Noise is the "static" in data — random, unpredictable ups and downs that have no specific cause. It is what remains after you remove the trend, seasonality, and cycles. Every real-world dataset contains noise.

Why It Matters: Avoiding "Over-Explaining"

Not every data fluctuation requires a policy response. If use-of-force incidents go from 12 to 14 to 11, this might simply be random chance — not a crisis or a success. Trying to "explain" noise leads to chasing ghosts, wasting resources, and confusing staff.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

Look at Precinct South throughout 2021–2023: the data oscillates between 22 and 28 with no clear direction. There is no trend. There is no strong seasonal pattern. It is bouncing around a stable average of approximately 25.

A manager who investigates every month's fluctuation in Precinct South is chasing noise, not signal.

How do you tell noise from a real signal? Ask:

4. Is the change larger than the typical month-to-month variation we normally see?
5. Does it persist for more than 2–3 months?